

OSAKA INTL, Japan

WMO: 477710

Lat: 34.784N

Lon: 135.439E

Elev: 15

StdP: 101.15

Time Zone: 9.00 (E09)

Period: 94-06

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.2	-0.2	-9.0	1.7	3.0	-7.2	2.1	3.8	9.1	7.0	8.3	6.2	2.4	320	0.434

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.3	34.9	25.6	33.9	25.2	32.9	25.0	26.6	32.1	26.2	31.4	25.7	30.7	4.0	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.1	20.3	28.9	24.9	20.0	28.8	24.1	19.1	28.6	83.3	32.3	81.3	31.3	79.4	30.7	28.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.4	7.4	6.5	-3.4	36.9	1.0	1.7	-4.1	38.1	-4.7	39.1	-5.3	40.0	-6.0	41.3		
			-5.0	27.7	1.1	0.5	-5.8	28.0	-6.4	28.3	-7.1	28.6	-7.8	28.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.6	5.0	5.5	8.7	14.8	19.8	23.6	27.9	29.1	25.2	18.9	12.8	7.4
	DBStd	8.81	2.17	2.51	3.00	3.54	2.37	2.34	2.41	1.80	2.93	3.01	3.16	2.59
	HDD10.0	449	156	129	62	5	0	0	0	0	0	0	9	89
	HDD18.3	1734	413	360	298	113	13	0	0	0	0	30	167	340
	CDD10.0	2864	1	2	22	150	304	408	556	593	454	274	92	8
	CDD18.3	1108	0	0	0	8	59	158	297	334	204	46	1	0
	CDH23.3	10651	0	0	0	36	305	1061	3312	3991	1741	204	2	0
	CDH26.7	4205	0	0	0	3	39	272	1422	1850	596	22	0	0
Wind	WSAvg	3.0	3.0	2.9	3.2	3.2	3.2	3.0	3.1	3.3	3.2	2.8	2.7	2.7
Precipitation	PrecAvg	1340	50	67	101	102	142	179	170	110	163	119	80	58
	PrecMax	1728	104	142	172	180	355	392	413	376	387	304	168	105
	PrecMin	732	14	21	36	26	32	58	31	20	44	28	19	9
	PrecStd	267	24	32	32	41	75	76	109	82	83	64	41	28
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.2	17.2	20.2	26.8	29.9	32.2	36.1	37.0	34.1	29.0	23.2	17.1
		MCWB	10.2	11.9	13.0	17.3	18.9	23.5	25.8	25.6	25.6	21.1	17.1	12.5
	2%	DB	12.1	14.0	18.0	24.1	27.8	30.8	34.9	35.1	32.9	27.0	21.2	15.1
		MCWB	8.2	8.7	11.6	15.9	18.3	22.4	25.7	25.5	24.2	20.4	16.0	10.6
	5%	DB	11.0	12.1	16.2	22.8	26.2	29.2	33.8	34.1	31.2	25.8	20.0	14.0
		MCWB	7.4	7.8	11.2	15.1	17.8	21.7	25.3	25.2	23.7	19.2	15.3	9.9
	10%	DB	9.9	10.8	14.8	21.0	25.0	28.1	32.2	33.1	30.0	24.1	18.2	12.8
		MCWB	6.4	7.1	10.0	15.1	17.5	21.3	25.0	25.1	23.4	18.6	14.1	8.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.9	12.5	15.2	20.4	22.7	25.6	27.1	27.2	26.5	23.5	18.7	14.1
		MCDB	13.8	15.8	17.6	23.3	25.5	29.7	33.0	32.5	31.3	25.9	20.7	15.9
	2%	WB	9.4	10.2	13.5	18.6	21.2	24.7	26.6	26.7	25.7	21.9	17.5	11.9
		MCDB	11.0	12.3	16.5	21.4	23.7	28.1	32.3	32.1	30.2	25.5	20.2	14.2
	5%	WB	7.7	8.9	12.2	17.1	20.0	24.0	26.1	26.4	25.2	20.6	16.3	10.2
		MCDB	10.3	11.4	15.6	20.6	23.2	27.3	31.5	31.6	29.5	24.1	19.1	13.1
	10%	WB	6.2	7.2	10.7	15.7	18.8	23.1	25.6	26.0	24.5	19.4	15.1	9.0
		MCDB	9.1	10.1	13.7	19.7	23.3	26.4	30.7	31.0	28.5	23.2	18.0	12.3
Mean Daily Temperature Range	5% DB	MDBR	7.9	8.5	9.2	9.9	9.6	7.9	7.5	8.3	8.0	8.9	9.2	8.6
		MCDBR	9.6	11.2	11.6	11.8	11.5	9.9	9.1	9.4	9.2	9.9	9.9	10.0
		MCWBR	7.0	7.5	7.6	6.1	5.4	4.4	3.2	3.2	3.7	4.8	6.3	6.5
	5% WB	MCDBR	8.5	8.9	9.4	8.8	7.9	6.8	8.0	8.1	7.9	7.9	8.5	8.8
		MCWBR	7.1	7.2	7.6	6.2	5.5	3.6	3.1	3.1	3.6	4.8	6.3	6.6
Clear-Sky Solar Irradiance	taub	0.417	0.461	0.525	0.558	0.563	0.584	0.585	0.551	0.503	0.461	0.432	0.412	
	taud	2.090	1.994	1.855	1.829	1.880	1.898	1.945	2.023	2.125	2.156	2.137	2.121	
	Ebn at Noon	752	763	747	747	750	732	727	745	761	757	732	729	
	Edh at Noon	126	155	194	208	200	196	186	170	146	131	120	116	
All-Sky Solar Radiation	RadAvg	2.16	2.80	3.82	4.71	5.18	4.70	4.96	5.09	4.00	3.22	2.41	1.97	
	RadStd	0.16	0.23	0.30	0.41	0.53	0.44	0.66	0.47	0.34	0.31	0.20	0.13	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page